3. Scale by $\sqrt{d_k}$

Here $d_k = 2 \rightarrow$ so $\sqrt{d_k} = \sqrt{2} \simeq 1.414$

$$S = \frac{QK^T}{\sqrt{2}} \approx \begin{bmatrix} 1/1.414, & 1/1.414 & 0/1.414 \\ 0/1.414, & 1/1.414 & 1/1.414 \\ 1/1.414, & 2/1.414 & 1/1.414 \end{bmatrix} \approx \begin{bmatrix} 0.707 & 0.707 & 0 \\ 0 & 0.707 & 0.707 \\ 0.707 & 1.414 & 0.707 \end{bmatrix}$$

S is the scaled score matrix

remember: for a row $[x_1, x_2, x_3]$ $\quad softmax(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$

$row_1 = [0.707 \; 0.707 \; 0]$

$e^{0.707} \approx 2.028$, $e^0 = 1$ $\rightarrow$ denominator $= 2.028 + 2.028 + 1 = 5.056$

$row_2 = [0 \; 0.707 \; 0.707] \rightarrow$ denominator $= 1 + 2.028 + 2.028 = 5.056$

$row_3 = [0.707, 1.414, 0.707] \rightarrow$ denominator $= 2.028 + 4.112 + 2.028 = 8.168$

$\alpha_{11} = 2.028/5.056 \approx 0.401$

$\alpha_{12} = 2.028/5.056 \simeq 0.401$ $\qquad \sigma_1 \approx [0.401, 0.401, 0.198]$

$\alpha_{13} = 1/5.056 \simeq 0.198$

Attention Weights row_2

$\sigma_2 \simeq [0.198, 0.401, 0.401]$

Attention Weights row_3

$\sigma_3 \simeq [2.028/8.168, 4.112/8.168, 2.028] \simeq \sigma_3 \simeq [0.248, 0.503, 0.248]$

Attention $\overset{\text{weight}}{\cancel{\text{matrix}}}$ is $\sigma = softmax(S) \approx \begin{bmatrix} 0.401 & 0.401 & 0.198 \\ 0.198 & 0.401 & 0.401 \\ 0.248 & 0.503 & 0.248 \end{bmatrix}$

Each row sums to ~1. Great!

Interpretation of Attention weight matrix

- Row 1 = token1 attends mostly ~~to~~ token 1 & 2 and less 3
- Row 2 = token2 attends less to 1 and mostly to 2 & 3
- Row 3 = Token3 attends most to token 2, equally less to 1 & 3